MILITARY SPECIFICATION

SAFE AND ARMING (CLEARING CHARGE TRAIN) ASSEMBLY FOR MINES, M718/M741 SYSTEM

This Specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This Specification contains requirements not normally covered by the drawings, and provides Quality Assurance provisions for the assembly of a safe and arming device for mines used in the M718/M741 System ballistic testing.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions General Specification For.
MIL-D-48120	-	Detonator, Electric: XM100; Parts for and Loading, Assembling and Packing.
MIL-D-63197	-	Detonator Delay Electric: PA506; Parts For, and Loading, Assembling and Packing.
MIL-L-63246	-	Lead Assemblies for S&A Mechanism For Scatterable Mines; Parts for, and Loading, Assembling and Packing.

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-453 - Inspection, Radiographic

DRAWINGS (See 6.7)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9317620 - Safe and Arming (Clearing Charge Train) Assembly

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations if available from the Superintendent of Documents, U.S. Government Printing Officer, Washington, DC 20402. Orders should specify “49 CFR 0-190 (lastest revision)”).

(Copies of specifications, standards, drawings, and publications listed above, or in the item detail specification, required by suppliers in connection with specification procurement functions, should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts and assemblies. The parts and assemblies shall comply with all requirements specified on drawings listed in paragraph 2.1, and associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Radiographic examination. The safe and arming device shall be examined for a fully non-armed condition in accordance with MIL-STD-453 and 4.5.4 and in accordance to Dwg. 9317620. Further radiographic requirements shall include clear definition of all explosive components as well as flexible cable assembly and slider lock. Proper explosive loading shall be verified.

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3.4 Explosive train function. The two (2) detonators, transfer lead and aft ejection charge assembly shall all function high order when the detonators are initiated in accordance with the procedure specified in 4.5.6.

3.4.1 Aft ejection charge assembly (Dwg. 9277858). The aft ejection charge explosive elements shall function causing the propellant to ignite, resulting in a minimum of one hundred and sixty thousandths (0.160) inch deformation in the aluminum test plate at temperatures between minus forty degrees centigrade (-40°C) to plus sixty degrees centigrade (+60°C).

3.5 Detonator short circuit resistance. The resistance of the short circuit across the detonators on the flexible cable shall be less than fifteen thousandth (0.015) ohms when tested in accordance with 4.5.5.2.

3.6 Data requirements. The contractor shall forward copies of the results of any of his examinations or tests to the designated address, as required by the contract (See 6.5).

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship - parts and assemblies. Parts and assemblies shall be devoid of burrs, chips, dirt, grease, and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated. Exterior surface coatings shall be continuous, except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection and standard quality assurance provisions. Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply, and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer, for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

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Part Description	Drawing	Quantity
Slider Assembly	9317618	10
Safe and Arming (Clearing Charge Train) Subassembly	9317620	10
Cable and Detonator Assembly	9277868	10
Safe and Arming (Clearing Charge Train) Assembly	9317620	35
Chamber Assembly	9277986	10
Aft Ejection Charge Assembly	9277858	25
Setback Pin Assembly	9317617	10

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Safe and Arming (Clearing Charge Train) Assembly for M73/M741 System		SHEET 1 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Slider Assembly (Dwg. No. 9317618)				
	Examination for defects	10		3.2	4.4.2.1
	Safe and Arming (Clearing Charge Train) Subassembly (Dwg. No. 9317620)				
	Examination for defects	10		3.2	4.4.2.2
	Non-Arming Test	10		3.2	4.5.1
	Non-Arming Test of centrifugal weights	10		3.2	4.5.3
	Arming Test	10		3.2	4.5.2
	Cable and Detonator Assembly (Dwg. No. 9277868)				
	Examination for defects	10		3.2	4.4.2.3
	Safe and Arming (Clearing Charge Train) Assembly (Dwg. No. 9317620)				
	Examination for defects	10		3.2	4.4.2.4
	Tests				
	Detonator Resistance	25		3.2	4.5.5.1
	Detonator Short Circuit	25		3.5	4.5.5.2
	Closure, Firing Pin Stake Push Off Test	5		3.2	4.5.7
NOTES:					

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63383(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Safe and Arming (Clearing Charge Train) Assembly for M718/M741 System				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Closure, Slider Stake Push Off Test	5		3.2	4.5.7
	Radiographic Examination	10		3.3	4.5.4
	Explosive Test	25		3.4	4.5.6
	Chamber Assembly (Dwg. No. 9277986)				
	Examination for defects	10		3.2	4.4.2.5
	Aft Ejection Charge Assembly (Dwg. No. 9277858)				
	Examination for defects	25		3.2	4.4.2.6
	Functioning test	25		3.4.1	4.5.6.2
	Setback Pin Assembly (Dwg. No. 9317617)				
	Examination for defects	10		3.2	4.4.2.7

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of safe and arming assemblies & sub-assemblies shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Explosive detonators, leads, and MDF from the same lot interfix number from one manufacturer.

c. Propellant from the same lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63383(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Slider Assembly				9317618
					NEXT HIGHER ASSEMBLY
					9317620
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Center charge lead assembly slug missing		100%	3.2	Visual
102	Transfer charge lead assembly missing, or assembled incorrectly		100%	3.2	Visual
103	Groove pin missing, or assembled incorrectly		100%	3.2	Visual
104	Lead holder transfer charge stake improper		100%	3.2	Visual
105	Lead holder missing		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63383(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Safe and Arming (Clearing Charge Train) Subassembly	1 OF 2	9317620(3)
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Spring, centrifugal lock missing		100%(1)	3.2	Visual/Manual
2	Holder, spring, missing or loose		100%	3.2	Visual/Manual
3	Lock, centrifugal missing		100%	3.2	Visual
4	Any spring missing or improperly positioned		100%	3.2	Visual/Manual
5	Presence of "G" weight		100%	3.2	Visual/Manual
6	Presence of ball		100%	3.2	Visual/Manual
7	Presence of "G" weight lock		100%	3.2	Visual/Manual
8	Presence of set back pin assembly		100%	3.2	Visual/Manual
9	Non-arming test		100%	3.2	4.5.1
10	Non-arming centrifugal lock test		100%(4)	3.2	4.5.3
Major					
101	Presence of more than one spring per cavity		100%	3.2	Visual/Manual
102	Staking missing or improper		100%	3.2	Visual
103	Presence of rear locks		100%	3.2	Visual
104	Presence of rear lock springs		100%	3.2	Visual
105	Any closure missing, loose or improperly staked		100%	3.2	Visual/Manual
106	Arming test		100% (2)(4)	3.2	4.5.2

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63383(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Safe and Arming (Clearing Charge Train) Subassembly				9317620(3)
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor 201	Evidence of poor workmanship Notes: (1) Manual inspection is defined as the allowable probing of the unit by means of an approved instrument which will not cause damage to the sample. (2) Test is to be performed prior to inserting rear lock assembly. (3) Parts list shall be in accordance with 4.5.1. (4) Arming test shall always precede the non-arming tests.		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63383(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Cable and Detonator Assembly				9277868
					NEXT HIGHER ASSEMBLY 9317620
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder improper		100%	3.2	Visual
102	Detonator placement incorrect		0.40%	3.2	Visual
103	Assembly height, maximum		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63383(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Safe and Arming (Clearing Charge Train) Assembly	1 OF 1	9317620
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Assembly in partially armed or armed state, missing components or improper explosive loading (radiographic)		100%	3.3	4.5.4
Major					
101	Interrupter free to move		100%	3.2	Note 4 of Dwg. No. 9317620
102	Chamber assembly improperly staked		100%	3.2	Visual
103	Grounding clip missing or improperly inserted		100%	3.2	Visual
104	Cable and detonator assembly not fully seated		100%	3.2	Visual
105	Solder improper (grounding clip tab to cable assembly)		100%	3.2	Visual/Manual
106	Presence of all closure discs		100%	3.2	Visual/Manual
107	Shield assembly missing, loose, not staked, or improperly installed		100%	3.2	Visual/Manual
108	Detonator resistance (after shield assy.)		100%	3.2	4.5.5.1
109	MDF damage	---	100%	3.2	Visual
110	Detonator not shorted (after shield assy.)	---	100%(1)	3.5	4.5.5.2
111	Cable spacer missing or improperly installed	---	100%(2)	3.2	Visual
112	Closure firing pin stake push off test	---	---	3.2	4.4.3.2/4.5.7
113	Closure, slider stake push off test	---	---	3.2	4.4.3.2/4.5.7
114	Explosive test (see para. 6.8)		0.40%	3.4	4.5.6
Minor					
201	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES: (1) Verify short circuit after resistance measurements have been taken and the short circuit has been replaced.
(2) Inspection must be made during assembly operation.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63383(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Chamber Assembly				9277986
					NEXT HIGHER ASSEMBLY 9317620
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Aft ejection charge assembly missing or loose		100%	3.2	Visual/Manual
102	Pin, locating missing or improperly installed		0.40%	3.2	Visual
103	Stake missing or improper		0.40%	3.2	Visual
104	Potting compound missing, improperly applied or on surfaces designed to be free of compound		100%	3.2	Visual/Manual
105	Crack or break in MDF		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Aft Ejection Assembly				9277858
					NEXT HIGHER ASSEMBLY 9277986
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Functioning (see para. 6.8)	---	---	3.4.1	4.4.3.1/4.5.6
Minor 201	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63383(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Setback Pin Assembly		9317617
			NEXT HIGHER ASSEMBLY 9317620

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Protrusion of stop pin		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables and below.

4.4.3.1 Aft ejection charge assembly. One hundred and twenty five (125) samples shall be randomly selected from each lot and tested as follows at the specified temperatures.

Temperature	-40°C ±3°C	+25°C ±3°C	+60°C ± 3°C
Quantity	20	85	20

The lot shall be rejected if the total of rejects from all temperatures exceeds the following requirements:

Criteria	Accept No.	Reject No.
M5 Propellant Initation or Dud	1	2
Dent Measurement	1	2
Blow Through	6	7

4.4.3.2 Stake push off test. A "Push off test" shall be performed on the front and rear housing closures after being staked to the housing, to check for integrity of staking. The test shall be performed on two samples each, prior to the start of each shift, and then performed on one sample each, after every two hours production and at the end of the shift, until three consecutive shifts have been accepted. Thereafter, the test may be performed on one sample each, prior to the start of each shift, and one sample each after four hours production and at the end of the shift. Failure of a test sample at any time will cause the test to revert to the initial test conditions until three consecutive shifts have been accepted.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is identified in the tables beginning with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for approval, in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6.3 herein.

4.5 Test methods and procedures

4.5.1 Non-Arming of safe and arming subassembly. The subassembly shall be tested for the specified non-arming requirements using the following parts as a minimum for the sample.

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Part No.	Nomenclature
9277786	Centrifugal locks
9277790	G-Weight
9277871	G-Weight lock
9278008	Housing
9317618	Slider assembly
9317617	Set back pin assembly
MS19060-4810	Ball bearing

Also included are all associated springs, locks and closures for the above assembly.

Optional parts are:

Closure firing pin, Part No. 9317591
Rear locks, Part No. 9278057
Slider, lock, Part No. 9278060
Rear lock springs, Part No. 9278109
And all associated closures.

Also included for the optional parts are all associated springs, locks and closures for the above assembly.

The test sample must not contain the staked disc closure, Part No. 9277812 for this test.

The slider shall be maintained in the correctly oriented position during the arming and non-arming tests. The Contractor, when necessary, may utilize an interim slider lock for test purposes.

The S&A assembly shall be tested as follows using approved spin and testing equipment.

The S&A, after secured nesting in the spin fixture, shall be spun up to the arming speed specified on the applicable drawing.

To verify that the set back pin is in the non-armed position, the contractor shall attempt to arm the “G” weight (Dwg 9277790), by applying the minimum arming force as stated on the applicable drawing. The S&A subassembly shall not arm. The force on the “G” weight shall then be released,

The set back pin assembly, (Dwg 9317617), shall be moved to the armed position and held in place. Arming force shall be as specified in drawing 9317620.

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The S&A, shall be spun to the non-arming speed as specified on the applicable drawing. The set back pin shall remain in the armed position. The contractor shall attempt to arm the “G” weight, (Dwg 9277790), in accordance with the minimum all-arm requirement as stated on the applicable drawing.

While complying with the above, neither the G-weight, Part No. 9277790, the ball bearing, Part No. MS-19060-4810 or the slider assembly, Part No. 9317618 shall move out of the safe state, i.e., into the armed position or in a position resting on the centrifugal locks. Duration of the time of spin for the test equipment shall be approved by the. technical agency of 6.5.

After or during this spin test, the sub-assembly shall be examined for evidence of arming either in or out of the test nest.

Any S&A failing to comply with any or all of the non-arming requirements shall be classed defective and removed from the lot. (Non-destructive test).

4.5.2 Arming of safe and arming subassembly. The subassembly, consisting of the parts specified in 4.5.1 shall be spun up to the arming speed specified in Dwg. 9317620 using approved test equipment. When at the required arming speed and the set back pin assembly, Part No. 9317617, in the armed position, the G-weight part No. 9277790, shall be moved into the armed position and the slider assembly, Part No. 9317618, shall move into the full arm position. Failure of the sample to comply with all of the specified requirements shall cause the sample to be classed defective and removed from the lot.

NOTE: Arming forces on the G-Weight, Dwg. No. 9277790, and Setback Pin Assembly, Dwg. No. 9317617, need not be rigorously controlled or measured during spin if it can be shown that the static force requirements are met.

The sample shall be disarmed at the end of the test by entering into further tests (See 4.5.3) or by other means as approved by the technical agency listed in 6.5. (Non-destructive test)

4.5.3 Non-arming of centrifugal locks (safe and arming sub-assembly). An armed test sample, using approved test equipment shall be spun at the non-arming speed specified in (Dwg. 9317620). While at this speed, the slider assembly, part No. 9317618, shall be moved into the non-armed position and the instrument performing this operation shall be withdrawn or removed, The slider assembly, without any physical restraint upon it shall not move into the full

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armed position but must remain in a non-armed position. This shall be indicated by the slider assembly coming to rest on top of the centrifugal locks Part No. 9277786.

The S&A shall be in the full unarmed (safe) position after all tests have been completed. With the exception of the set back pin non-arming test specified in paragraph 4.5.1, (which may be performed during the arming sequence for convenience) all non-arming tests shall be performed after the arming tests. The order of the tests within the above arming and non-arming paragraphs is optional. Any S&A failing to comply with any or all of the requirements shall be classed defective and removed from the lot. (Non-destructive tests).

4.5.4 Radiographic examination. - The safe and arming assemblies radiographic plates shall be of a quality such that the Technical Agency of 6.5 or their appointed representatives shall be able to view all of the requirements to the Agency's satisfaction. The radiographic plates shall clearly define all explosive components; positively show that the assembly is in the full non-armed rendition; the flexible cable shall not be pinched, twisted or damaged; the MDF assembly shall not show evidence of any damage and all components shall be in the non armed position. Any assembly failing to comply with all of the specified requirements shall be classed defective and removed from the lot. (Non-destructive test).

4.5.5 Resistance measurements

4.5.5.1 Detonator continuity resistance. The resistance of the detonator as installed on the flexible cable and safe and arming assembly shall be measured for the value specified in (Dwg. 9317620). An approved millohmmeter whose excitation current does not exceed the no fire condition of the detonators shall be used. The resistance measurements shall be obtained across the eyelets on the flexible cable (Dwg. 9277774). Any assembly exhibiting resistance not conforming to the specified requirement shall be classed defective. The exposed conductor in the flexible cable (Dwg 9277774) must be cut to perform this test and soldered to reconnect the above conductor after completion of test. (Non-destructive test).

Any assembly failing to comply with the specified resistance requirements shall be classed defective. (Non-destructive test.)

4.5.5.2 Detonator short circuit resistance. The soldered flexible cable assembly shall be measured for the specified resistance using an approved milliohmmeter whose excitation current does

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not exceed the no fire condition of detonators across the eyelets. This test must be performed after the device passes the conditions specified in 4.5.5.1.

Any assembly failing to comply with the specified resistance requirements shall be classed defective. (Non-destructive test.)

4.5.6 In-line explosive functioning

4.5.6.1 Explosive train. The armed safe and arming device shall be tested in an explosion proof test chamber. (To accomplish the test below the soldered short on the flexible cable must be removed.) An eighty three (83) microfarad plus or minus five percent (± 5%) capacitor charged to five plus or minus one tenth (5.0 ± 0.1) volts direct current shall be applied to the cable inputs which will cause the assembly to explosively function. A visual examination shall be made of the assembly to ascertain if all components have also functioned. Any sample failing to comply with all of the specified test requirements shall be classed defective. (Destructive test)

NOTE 1: The explosive train performance test may be performed on assemblies consisting of all explosive elements, housing, slider assemblies, slider locks and staked shield assembly assembled in the unarmed condition, and then armed remotely. The contractor shall describe the proposed method of remote arming (see para. 4.4.4). All explosive elements and metal parts shall be a representative sample drawn from each lot of parts that will constitute one lot of safe and arming assemblies (i.e.; complying with the lotting provisions of 4.4.1).

4.5.6.2 Aft election charge assembly. The aft ejection charge testing shall be performed in an approved chamber, as per figures 1 thru 6 for the required functioning and dent criteria. The test plate shall be in accordance with the requirements shown in the specification drawing, figure 4. Initiation of the aft ejection charge assembly shall be performed in an approved manner using a M100 electric detonator (Ref MIL-D-48120) or equivalent.

Any aft ejection charge assembly failing to comply with all of the specified requirements shall be classed defective. (Destructive test).

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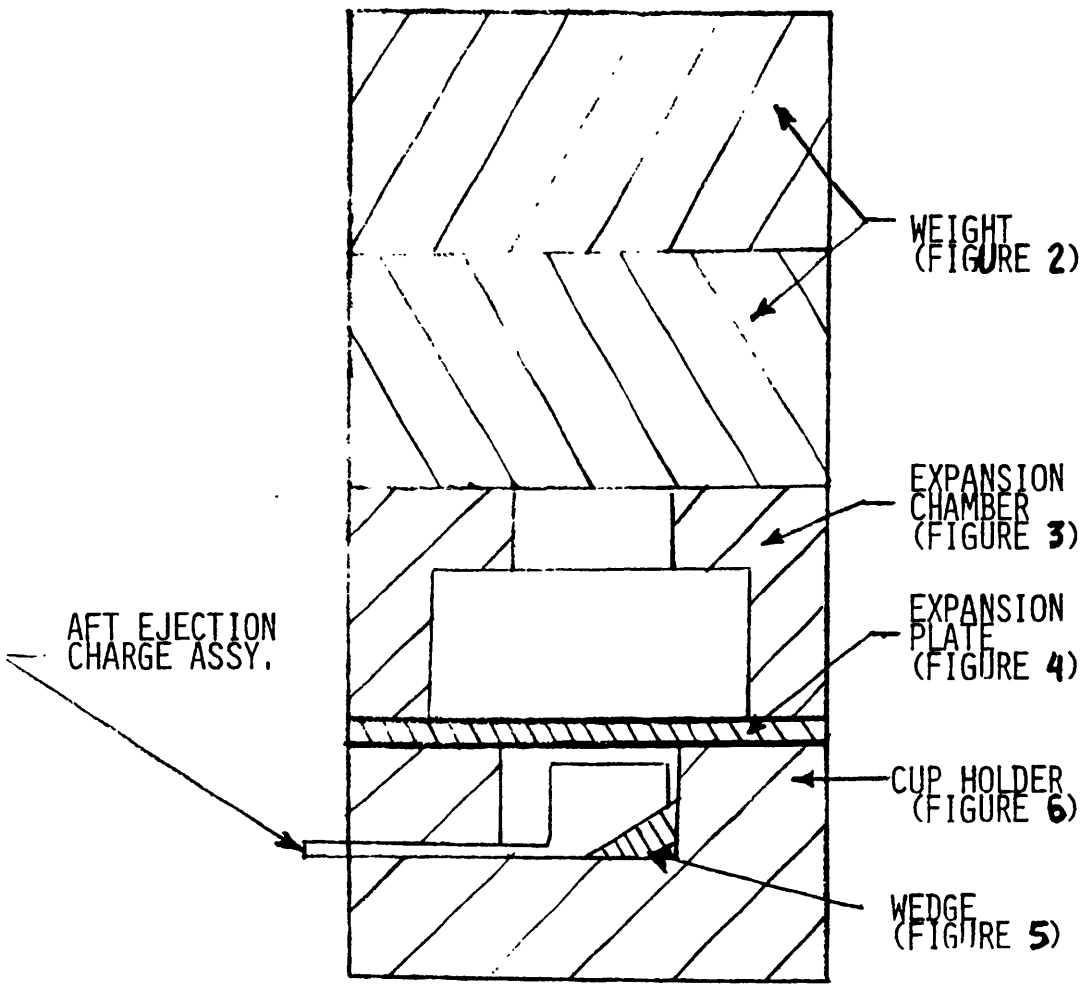
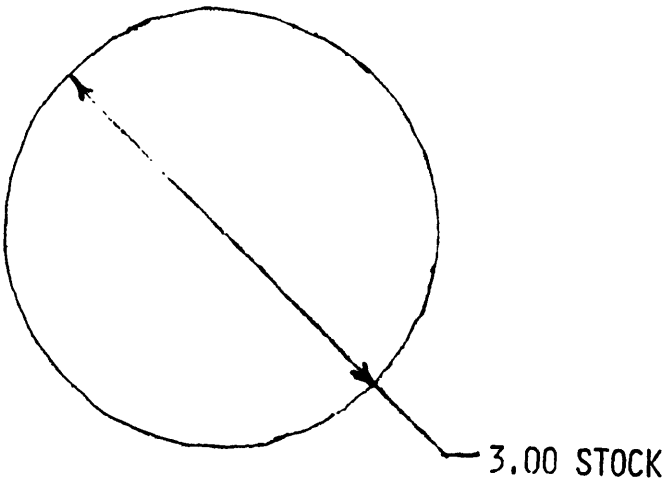
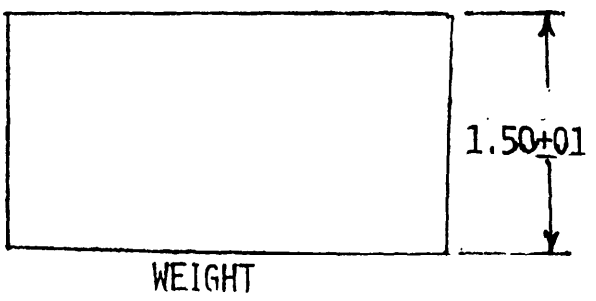


FIGURE 1

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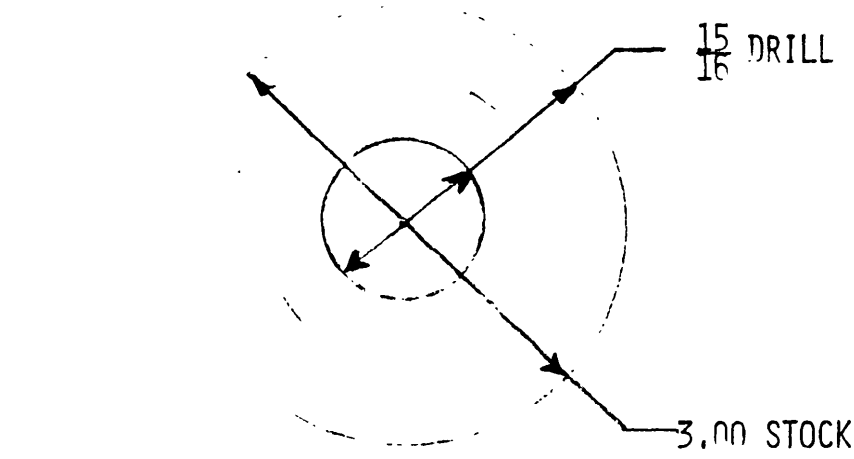


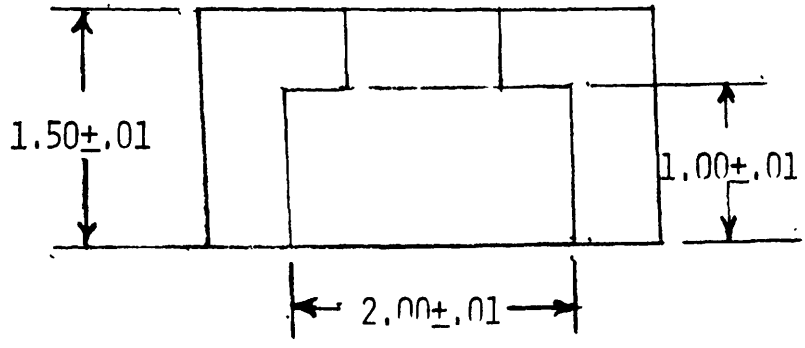


MATERIAL:
STEEL 1018
OR EQUIV.

FIGURE 2

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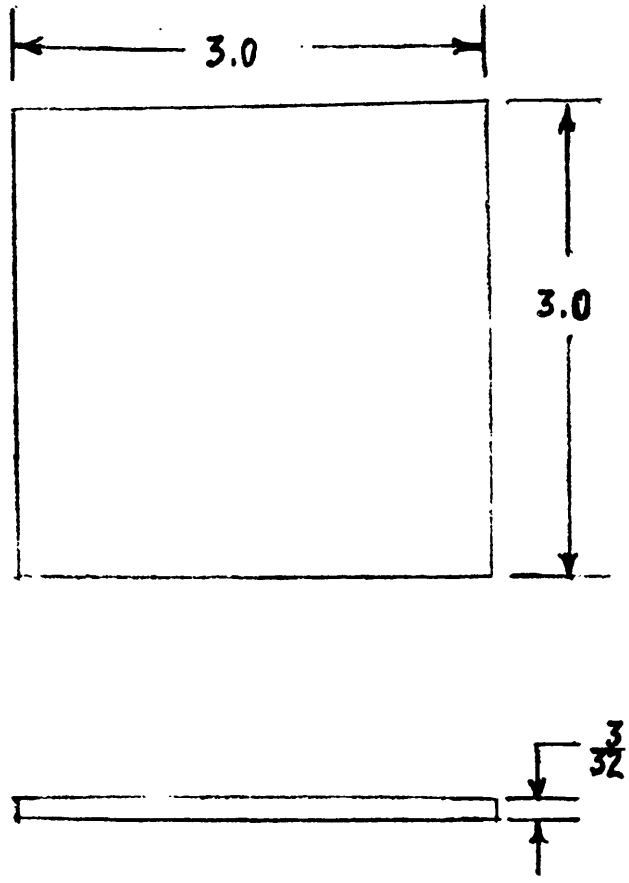


MATERIAL: 1018 STEEL
OR EQUIV.

EXPANSION CHAMBER

FIGURE 3

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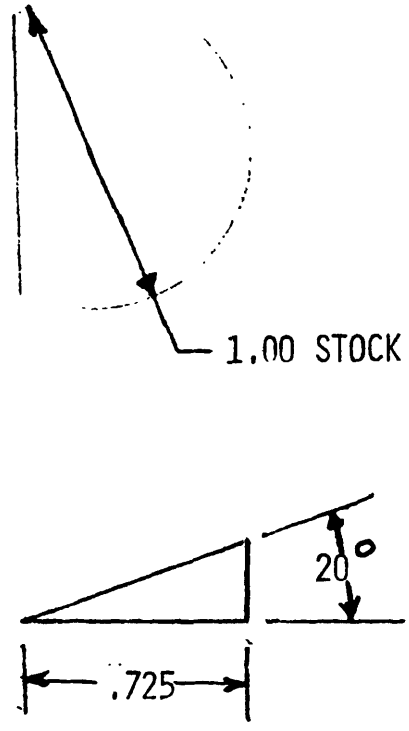


MATERIAL: ALUMINUM 6061-T6

EXPANSION PLATE

FIGURE 4

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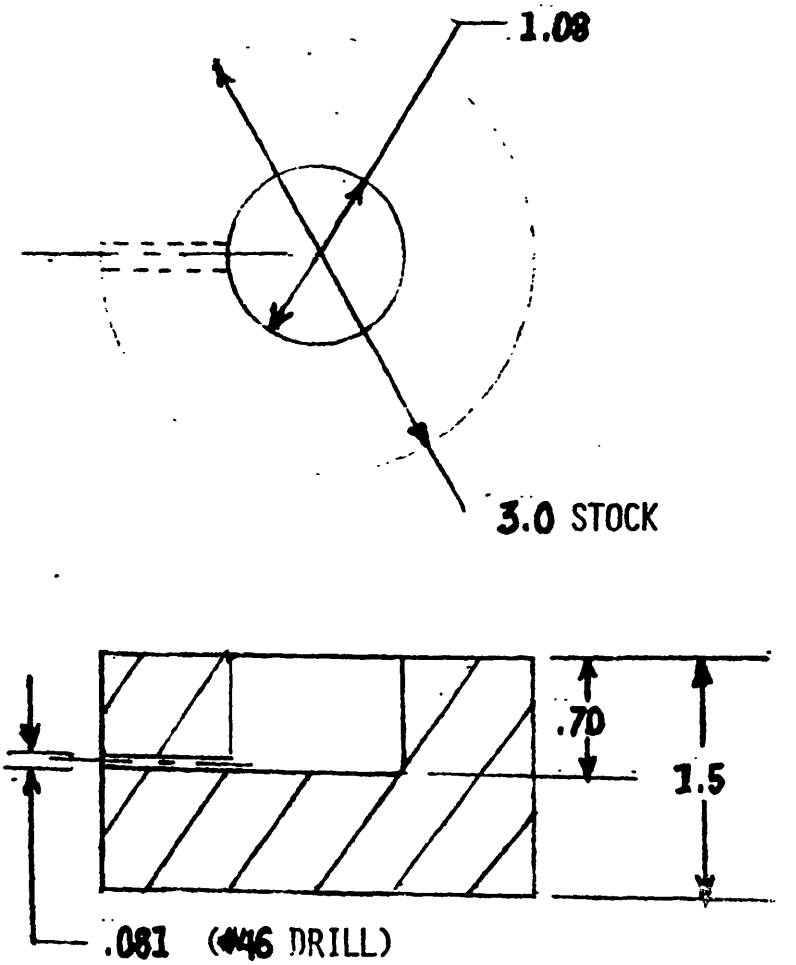


MATERIAL: 1018 STEEL
OR EQUIV.

WEDGE

FIGURE 5

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MATERIAL: 1018 STEEL
OR EQUIV.

CUP HOLDER

FIGURE 6

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4.5.7 Stake push off test. The test shall be performed using approved test equipment on samples of housings and closures staked in accordance with the specified requirements and with no components within the housing. If any sample fails to comply with all of the specified requirements it shall be classed defective and all production items preceding the last successful push test shall be rejected. (Destructive test).

4.5.8 Check test for deterioration of detonators and leads. (See MIL-L-63246, MIL-D-63197 and MIL-D-48120). If the total time between original acceptance of any detonator or lead lot and the assembly of the lot into the safe amd arming device exceeds two years, or if the detonators or leads have been subjected to adverse conditions, however brief, at any time since previous' tests, the detonator or lead lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-L-63246 and/or MIL-D-63197, and/or MIL-D-48120. Immediately before the detonator or lead lot is assembled, the Government Inspector at the facility assembling the detonators or leads into the safe and arming device shall select the sample for deterioration tests and the contractor shall verify that the devices comply with the requirements of the above specifications.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level - Commerical. The safe and arming device shall be preserved and packaged utilizing industrial procedures and in accordance with Code of Federal Regulations Title 49 100-199. These procedures shall be submitted for examination in accordance with Para. 6.5.

5.2 Packing

5.2.1 Level - Commerical. The safe and arming device preserved and packaged as specified in 5.1.1 shall be packed for shipment in accordance with acceptable industrial procedures such as to prevent deterioration or damage during handling and shipment to the first destination. Industrial packing procedures shall be in accordance with Code of Federal Regulations Title 49 100-199

5.3 Marking. Marking shall be in accordance with MIL-STD-129 and Code of Federal Regulations Title 49 100-199.

5.4 Shipping. When shipments from more than one lot are shipped, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Intended use. The assembly covered by this specification is intended for use in the Projectile, 155mm, AT, M718/M741 special ballistic test items.

6.2 Ordering data. See MIL-STD-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, Attn: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Cost of check test. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the detonators and/or leads. The tests shall be conducted at Government expense without cost to the contractor who loaded the detonators and/or leads or to the contractor assembling them into the Safe and Arming device. This shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., while the lot of detonators and/or leads was under the jurisdiction of either contractor.

6.5 Submission of results of contractor conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARRADCOM, Attn: DRDAR-QAA-M.

6.6 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, ARRADCOM, Attn: DRDAR-QAA-M, Dover, New Jersey 07801.

6.7 Drawings. Drawings listed in Section 2 of this Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under cognizance of ARRADCOM.

6.8 Explosive test. The contractor may request a waiver of the explosive test provided the following criteria are met:

(a) The contractor is currently producing Safe and Arming Devices for Mines, M718/M741 Systems, P/N 9278015.

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(b) The metal parts, detonators, cable and detonator assemblies, transfer lead, and aft ejection charge assemblies are from the same lots of components used to produce a lot of P/N 9278015 S&A's.

(c) The above lot of 9278015 S&A's successfully passes the explosive train test specified in MIL-S-48747A.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1345-A-127

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-63383 Safe and Arming (Clearing Charge Train) Assembly for Mines, M718, M741 System

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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